Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/31/2006
PI Name:	Vazquez, Marcelo Ph.D.		
Project Title:	Risk Assessment and Chemoprevention of HZE-Induced CNS Damage		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Radiation Effects Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	11973	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	05/01/2001	**End Date:**	04/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	RE00202		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Because successful operations in space depend on the performance capabilities of astronauts, radiation-induced neurological damage, could jeopardize the successful completion of mission requirements, as well as have long-term consequences on the health of astronauts. It is therefore necessary to understanding the nature of this risk in order to assess its seriousness and to develop countermeasures. Compared to the large literature associated with radiation therapy, knowledge is limited about the cellular and molecular responses of cells to high-LET HZE radiation in general, and very limited about the central nervous system (CNS) specifically. Therefore, we propose to compare the effects of charged particle (Fe, Si), protons, gamma and x-ray radiation on the cells of the CNS, namely neurons and glial (astrocytes and oligodendrocytes). Cell cultures of CNS cells, both cycling and postmitotic differentiated cells, will be utilized as model systems. We will test the hypothesis that exposure to low fluences/doses of heavy ions and protons can induce cell death in neural CNS neural cells and that increasingly dense ionizing radiation will be increasingly toxic. The activation of two separate stress signal transduction pathways will be examined (p53 and ceramide) for their role in causing cell death or		

	other deleterious changes caused by irradiation. And with respect to p53, we will determine which of the post-translational modifications in regulating p53 function are relevant for charged particle induced cell death. Finally, we will will test the hypothesis that modulating the stress signal transduction pathways will modify the radiation response of brain cells exposed to heavy ions and protons, and test the efficacy several compounds as potential countermesures for HZE radiation toxicity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research is related to the effects on humans of space radiation with the ultimate goal of providing a firm scientific basis for space radiation protection. These experiments address the particular problem of high-energy heavy ion radiation exposures during future long-term deep space flights. The principal objective is to improve our understanding of the neurobiological effects of low fluences of densely ionizing charged particles on brain structures. In addition, these studies have the potential for enhancing and extending our knowledge of the structure and reparability on genetic material, and understanding the link between ionizing radiation and biological effects such as mutagenesis, neurodegeneration, and aging. Although the radiation fields encountered in space are both quantitatively and qualitatively different from those on Earth, comprehension of their mechanisms of action will increase understanding of basic mechanisms of action of ionizing radiation, and the use of high energy charged particles in radiotherapy. If drugs prove successful in reducing radiation effects from low doses of highly ionizing radiations, this research may lead to improved responses for patients undergoing radiation therapy for brain tumors
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated:)